

ILIAMNA, AK, USA

WMO: 703400

Lat: 59.749N

Lon: 154.909W

Elev: 143

StdP: 14.62

Time Zone: -9.00 (NAK)

Period: 94-19

WBAN: 25506

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-18.4	-12.9	-26.6	1.3	-16.5	-21.7	1.7	-11.5	34.1	35.7	31.4	27.8	9.1	350	0.770

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.3	72.2	59.1	68.3	57.5	64.9	55.7	60.9	70.1	58.8	66.1	57.1	62.9	9.4	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
57.0	69.6	63.0	55.3	65.4	61.4	54.0	62.2	59.3	26.9	69.9	25.5	66.2	24.5	62.9	68.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 28.8	25.5	21.9	-19.6	78.6	6.8	4.3	-24.5	81.7	-28.5	84.3	-32.4	86.7	-37.4	89.8		
(5)			-19.4	63.0	5.9	2.7	-23.6	65.0	-27.0	66.6	-30.3	68.1	-34.6	70.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	36.7	17.8	23.4	22.7	34.4	44.4	52.2	56.2	55.4	48.7	37.5	25.9	20.5		
	DBStd	17.27	17.46	14.11	12.32	7.41	5.45	4.55	4.48	4.21	4.78	8.45	12.40	15.43		
	HDD50	5378	998	745	845	469	188	24	2	5	76	388	723	915		
	HDD65	10351	1463	1165	1310	919	639	385	277	298	490	852	1173	1380		
	CDD50	510	0	0	0	0	14	89	195	174	36	2	0	0		
	CDD65	6	0	0	0	0	0	1	4	1	0	0	0	0		
	CDH74	65	0	0	0	0	2	18	40	5	0	0	0	0		
	CDH80	7	0	0	0	0	0	2	5	0	0	0	0	0		
	WSAvg	9.7	11.1	10.5	9.1	9.2	9.1	8.3	8.2	8.2	9.3	10.4	11.1	11.7		
	PrecAvg	28.0	1.4	1.1	1.3	1.3	1.5	1.9	2.8	4.8	4.5	3.3	2.2	1.8		
(15) Precipitation	PrecMax	40.8	2.8	2.3	4.5	2.6	4.1	4.3	6.6	10.5	11.0	8.0	7.9	4.6		
	PrecMin	15.3	0.3	0.2	0.0	0.2	0.0	0.3	0.2	2.0	1.5	0.2	0.1	0.0		
	PrecStd	7.0	0.7	0.6	1.1	0.8	0.9	1.2	1.8	2.2	2.6	1.7	1.8	1.1		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	45.3	43.7	45.9	55.2	69.5	76.7	79.1	74.3	64.2	54.3	49.0	45.3		
		MCWB	39.3	38.4	39.4	44.1	52.9	60.1	63.2	61.4	53.9	47.7	43.8	41.0		
	2%	DB	42.3	41.2	42.0	50.4	62.9	70.4	72.8	69.6	60.1	52.1	46.7	43.1		
		MCWB	38.3	37.2	37.3	41.5	49.5	56.4	60.5	59.5	52.4	47.1	42.5	38.9		
	5%	DB	40.3	39.3	39.4	47.2	58.3	65.6	68.5	66.2	57.7	50.3	44.7	41.2		
		MCWB	37.2	36.0	35.6	40.0	47.6	54.6	58.6	57.5	51.5	45.9	41.0	37.5		
	10%	DB	38.6	37.7	37.4	44.3	54.0	61.9	64.6	63.0	55.7	48.4	42.0	39.1		
		MCWB	35.9	34.7	34.2	38.5	45.5	52.7	56.6	55.9	50.8	44.4	38.9	36.1		
	0.4%	WB	41.1	39.4	40.5	45.5	54.3	61.4	64.7	63.6	56.0	50.2	45.2	41.9		
		MCDB	44.3	42.3	44.4	54.0	67.0	74.4	76.2	70.6	60.8	52.7	47.5	44.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	38.8	37.8	38.1	42.3	50.6	58.0	61.5	60.6	54.2	48.0	43.5	39.9		
		MCDB	41.5	40.5	41.3	48.9	60.7	67.8	71.8	67.6	57.7	51.1	46.4	42.7		
	5%	WB	37.4	36.4	36.1	40.5	48.2	55.5	59.2	58.5	52.8	46.3	41.2	37.7		
		MCDB	39.9	38.9	38.8	46.2	57.1	63.7	67.1	64.4	56.1	49.6	44.3	40.3		
	10%	WB	36.0	35.0	34.6	39.0	46.2	53.3	57.5	56.9	51.4	44.9	38.9	36.2		
		MCDB	38.5	37.3	37.1	43.6	53.1	60.4	63.3	61.5	54.8	48.2	41.8	38.8		
	5% DB	MDBR	11.1	11.6	13.9	13.1	14.4	14.8	12.3	11.8	11.2	10.2	10.6	11.1		
		MCDBR	8.7	8.4	11.3	17.3	23.5	23.3	20.7	18.2	14.4	9.4	8.1	8.3		
		MCWBR	6.4	6.2	8.0	10.5	12.8	11.6	10.2	9.7	8.5	6.4	6.0	6.4		
		MCWBR	8.0	8.1	9.6	15.4	20.9	20.8	18.7	15.7	11.6	7.9	8.2	8.3		
(40) Clear-Sky Solar Irradiance	taub		0.245	0.283	0.306	0.385	0.383	0.376	0.374	0.358	0.326	0.302	0.260	0.246		
		taud	2.137	2.244	2.208	2.066	2.259	2.365	2.420	2.458	2.523	2.415	2.098	1.956		
	Ebn at Noon		173	231	261	254	267	272	269	266	257	224	157	110		
		Edh at Noon	14	22	31	43	38	35	32	29	23	18	14	11		
(44) All-Sky Solar Radiation	RadAvg		116	307	824	1264	1514	1669	1402	1104	755	384	156	69		
	RadStd		17	61	128	154	164	142	127	141	110	34	23	6		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	+5.69	+5.00	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (11 neighbors)		N/A	+4.32	+3.46	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page